

CHERUIYOT BUSES KIPNG'ENO
GEOSPATIAL DATA SCIENTIST | SPATIAL INFORMATION MANAGEMENT |
PYTHON | REMOTE SENSING | MACHINE LEARNING

Email: cherukip@gmail.com |  +254 720 347 328

PROFILE SUMMARY

Results-driven Geospatial Data Scientist with 12+ years of experience designing and delivering spatial analytics solutions across donor-funded programs, research and development planning. Strong background in GIS, Remote sensing, Python automation, machine learning, and geospatial decision-support systems. Proven experience translating complex spatial datasets into practical planning recommendations for governments, NGOs, and private-sector stakeholders. Experienced in managing and working with multicultural teams and client-facing project delivery. Passionate about advancing inclusive and sustainable service access.

RELEVANT TOOLS AND SKILLS

- **Programming and Data Analysis:** Python, SQL, R
- **Python Libraries:** Pandas, NumPy, Scikit-learn, GeoPandas, Rasterio, Shapely
- **GIS:** ArcGIS Pro, QGIS, PostGIS, Python
- **Remote Sensing:** Google Earth Engine, Sentinel, Landsat, SNAP, QGIS
- **Data Visualization:** Power BI, Excel, Tableau
- **Databases:** PostgreSQL, Spatial Databases

EDUCATIONAL BACKGROUND

MSc in GIS & Remote Sensing – Jomo Kenyatta University of Agriculture & Technology

- Thesis: *Mapping Flood Vulnerability in Budalangi, Western Kenya*

BSc in Environmental Science – Kenyatta University

Professional Trainings:

- Digital Technology and Artificial Intelligence for Climate Action Course- African Group of Negotiators Expert Support (AGNES)- Jan – April 2026.
- Python for Machine Learning & Data Science Masterclass – Udemy – 2026.
- Generative AI Course – Open University of Kenya and BCS Technology- 2025
- Introduction to Research Ethics, TRREE Training Program in Research Ethics and AMREF- 2022.
- Introduction of python for Data Science- 2018- Code Academy.
- Open data in Agriculture and Nutrition – 2017- GODAN Action & UKAID.
- Building Web application using ArcGIS API for JavaScript – 2014- ESRI EAST Africa.

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KEY ACHIEVEMENTS

- Conducted qualitative research on child trafficking in Kisumu and Kakamega counties, engaging criminal justice actors and community stakeholders to analyse trafficking patterns and response mechanisms.
- Designed and implemented large-scale household surveys (5,000+ households), including tool development, sampling, and data quality assurance for Water Insecurity study in Turkana County - 2023
- Conducted several FGDs and KIs to gather insights on socio-economic, environmental, and community resilience issues for different clients
- Established and managed **GIS systems for two organizations** (World Vision and DAI Global) supporting spatial analysis, data integration, and evidence-based research and resources allocation and planning.
- Successfully supported planning, monitoring and evaluation of multi-million-shilling development projects in World Vision, DAI Global and other consulted projects.
- Improved data quality and reporting accuracy through structured M&E systems.
- Been engaged in data collection, analysis, and reporting for more than 20 donor funded projects to support in planning and resources allocation
- Engaged in various GIS data analysis, visualization and reporting
- Successfully, with other team members developed land use plans and invasive species management plans for Baringo, Isiolo, Samburu and Laikipia counties under RangerER project – Year 2025.

PROFESSIONAL MEMBERSHIP

- Member, Environmental Institute of Kenya – Member No – EIK/I/I035

PROFESSIONAL EXPERIENCE

I. Freelance Consultant (GIS & Remote Sensing) – Kenya | Various Assignments
Feb 2025 – Present

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- Conducted qualitative data collection (FGDs and KIs) for the study into Child Trafficking in Kisumu and Kakamega Counties for the Child Trafficking study in 14 Counties in Kenya under the National Crime Research Centre
- Led the collection of quantitative and qualitative data collection, analysis and report drafting for the NEMA Adaptation fund End of Project Evaluation in Machakos, Laikipia, Marsabit, Homabay, Kisumu, and Kwale Counties
- Produced **high-quality analytical reports**, maps, and visualizations translating complex datasets into actionable insights for decision-makers for different individual clients
- As GIS Consultant in land degradation mapping and development of Land degradation management plans in Baringo, Samburu, Isiolo and Laikipia Counties under EU-funded RangER Project, I collected Ground Truth data and processed satellite images to map the land degradation
- GIS Consultant in Invasive species mapping and development of invasive species management plan in Laikipia, Isiolo, Samburu and Baringo Counties.

2. Quantitative Research Fellow – REACH WISER Project, Univ. of Nairobi & Univ. of Oxford

Jan 2023 – Jan 2025

- Managed large-scale household surveys (5,000+ households) on socio-economic and water insecurity factors and climate risk and resilience projects. Successfully designed the data collection tools for SurveyCTO data collection, Trained enumerators, piloted and supervised data collection and managed the Survey CTO data platform
- Developed annual project workplans programs and project budget.
- Conducted FGDs and KIs in Turkana County on climate risk and community resilience, to supplement the household's data
- Analysed the household's data, prepared research outputs, and co-authored study reports and disseminated findings in different forums.
- **Developed personal research on the Impact of Flooding on water security in Turkana Central: Analysed satellite images to map flood zones and water resources and population at risk of flooding disruption. The paper is on review for publication.**

3. GIS & Remote Sensing Consultant – JKUAT Enterprises (RangER Project)

Jul 2022 – Mar 2023

- Conducted **land use/land cover mapping and ecological assessments** to support community conservancy planning in Samburu, Baringo, Laikipia and Isiolo Counties.

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- Organized **community meetings, FGDs, and KIIs** to collect socio-economic data for land-use planning.
- Produced **spatial and statistical analyses** linking socio-economic activities to conservation and security challenges.
- Codeveloped the Land use plans for Conservancies in Baringo, Isiolo, Samburu and Laikipia Counties

4. GIS Research Consultant – AgriChains Ltd

Jul 2020 – Jun 2022

- Led the GIS and remote sensing data collection and analysis for different clients and organizations. Additionally produced projects reports.
- Led data collection, household surveys, and KIIs in sanitation and WASH projects.
- Produced research reports, infographics, and policy briefs for different stakeholders and consultancies.

5. GIS Specialist – Development Alternatives Inc. (USAID KIWASH Project)

Feb 2016 – Jun 2020

- Designed and managed national GIS datasets on water, sanitation, agriculture, and natural resources for the KIWASH project
- Led GIS analysis for water access and infrastructure planning in underserved communities.
- Trained Water Service providers on utility projects mapping, water pipeline mapping and reduction on Non Revenue Water utilizing GIS technology.
- Developed spatial decision-support tools for prioritizing investments and service expansion.
- Integrated demographic, environmental, and infrastructure datasets into planning systems.

6. GIS Officer – World Vision International Kenya

Jan 2013 – Jan 2016

- Conducted Energy Resources Mapping for the Renewable Energy Resources Management project.
- Designed and managed the GIS database system for World Vision Kenya. Additionally led the collection of projects GIS data, designed the data collection tools, performed various spatial analysis and provided GIS products and resources to support programming.
- Supported research in health, energy, education, and community development projects through GIS and data analytics.
- Conducted disease surveillance mapping (malnutrition and malaria) to support health interventions.
- Developed digital household surveys and baselines for project monitoring and evaluation.

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- Facilitated capacity building for staff on data collection, analysis, and reporting.

7. Asals Envirotech Ltd. Consultants-Nairobi (2010-2012)

Position - GIS and EIA Research Officer

- GIS expert in carrying out “WASH (Water, Sanitation and Hygiene) project resources and Facilities Mapping”, World Vision Kenya, Bartabwa Integrated Program Area, August 2012.
- GIS expert in carrying out “WASH (Water, Sanitation and Hygiene) project resources and Facilities Mapping”, World Vision Kenya, Nyakach Integrated Program Area, June 2012.
- GIS assistant in carrying out “WASH (Water, Sanitation and Hygiene) project resources and Facilities Mapping”, World Vision Kenya, Mutomo Integrated Program Area, February 2012.
- GIS and Remote Sensing Associate Expert in carrying out GIS database development for the Ministry of special programs Kenya- Western Kenya Flood Mitigation Community Driven Development project, 2010
- GIS data collection, processing, data visualization, Digital image processing and remote sensing data analysis for various applications as per clients’ requirements.
- Participates in regular geospatial data collection, processing, analysis and presentation for several clients.
- Development of a very high spatial resolution digital elevation model of Western Kenya (Specifically River Nzoia basin for flood mitigation projects) under the Western Kenya Flood Mitigation Project, base on Lidar acquired CAD data points.
- Development of informative maps and information for communication to clients and other parties based on geospatial data analysis and processing.
- Environmental Impact Assessment (EIA) Associate Expert in carrying out the Environmental Impact study for the proposed 6 community boreholes in Mwala IPA, funded by World Vision, Mwala Integrated Program Area-IPA, September 2012
- Environmental Impact Assessment (EIA) Associate Expert in carrying out the Environmental Impact study for the proposed Magoi Community Waterpan in Mogotio, funded by World Vision, Mogotio Integrated Program Area-IPA, April 2012
- EIA Associate Expert in undertaking Environmental Impact Analysis for the proposed four boreholes in Korr, Ngurunit, Illaut and Ulauli funded by World Vision Kenya in Laisamis Integrated Program Area-IPA, Marsabit South District . March 2012.
- EIA Associate Expert in undertaking Environmental Impact Analysis for the proposed Kwa

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Kamwana, Katitu, Mkolosoni and Kawandei boreholes in Mutomo Integrated Program Area-IPA. March 2012.

- EIA Associate Expert in undertaking Environmental Impact Analysis for the proposed World Vision offices in Dadaab. December 2011.
- EIA Associate Expert in undertaking Environmental Impact Analysis for the proposed Eleven Falls Service Filling Station in Kivaa Market, Masinga District. August 2011
- EIA Associate Expert in undertaking Environmental Impact Analysis for the proposed Kwa Kamwana community sand harvesting projects in Masinga District. August 2011
- EIA Associate Expert in undertaking Environmental Impact Analysis for the proposed rock water catchment in Ngurnit cluster, Laisamis. World Vision Kenya, Laisamis Integrated Programme Area. July 2011
- EIA Associate Expert in undertaking Environmental Impact Analysis for the proposed 3 boreholes in three different schools in Nyakach, Nyando District. World Vision Kenya, Katito Integrated Programme Area. June 2011
- Matete Integrated Programme Area. May 2011.

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REFEREES

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